

PHYSICAL AI HACKATHON

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PROBLEM & CHALLENGES

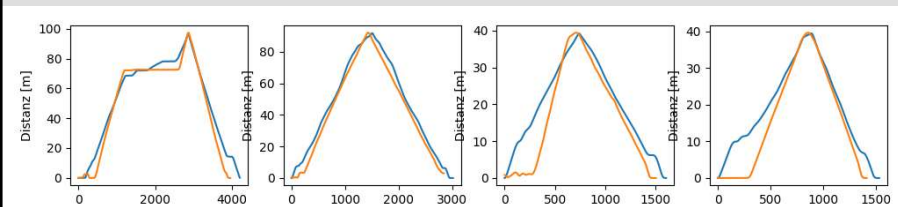
Content

- Estimate camera position inside a pipe from video frames
- Detect turning point (forward → backward)

Challenges

- Water, reflections, sludge
- Low texture / poor visibility
- Overexposure and blur
- Intro/logo frames at the beginning
- Varying pipe lengths

MOVEMENT ESTIMATION APPROACH



Content

- Optical Flow (Lucas-Kanade)
→ track feature points across frames
- Integrate motion → position curve

Turning Point Detection

- Plateau detection on position curve
- Use midpoint of plateau

Stabilization

- Triangle-wave prior (LOOCV-based)
- Weighted combination:
Flow + Prior
- Smoothing + monotonic constraints

BONUS: CONDITION ANALYSIS

Content:

- Lightweight computer vision approach
(no heavy ML models)

Pre-framed features

- Visibility score
- Blockage degree
- Obstruction type
- Dirty frame ranges

Features used:

- Brightness & contrast
- Canny edge density
- HSV color statistics
- Center-region color analysis
- Texture / edge patterns

BONUS OUTPUTS AND INTERPRETATION

Content

- Detects poor visibility from darkness, overexposure, or low contrast

Blockage Degree

- Edge density used as obstruction proxy
- High values indicate dirty / obstructed regions

Obstruction Types

- Gravel
- Lime
- Roots
- Sludge
- Unknown

OUTPUTS & INSIGHTS

Content

- Risk score over time
- Visibility timeline
- Condition timeline
- Debug frames & anomaly detection

Key Idea

Pipe Health
Timeline

Example:

- Intro → low visibility
- Water → medium risk
- Roots → high risk

Takeaway

- Robust movement estimation
- Interpretable condition analysis
- Practical inspection insights